

Involution Palindrome DNA Languages *

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Abstract

This paper aims to investigate properties of involution palindrome languages. Let θ be either a morphic or antimorphic involution. we show that if θ is an antimorphic involution, then the set of all θ -palindrome words is a proper subset of all skew θ -palindrome words. We give a characterization for that a word is skew θ -palindrome. That is, A word is skew θ -palindrome if and only if it is a product of two θ -palindrome words. We show that if a word w is skew θ -palindrome, then w^n is skew θ -palindrome for every $n \geq 2$. We also give a characterization for that a word is a non-involution palindrome.

Keywords: Involution Palindrome, Skew Involution Palindrome, DNA Languages

1 Introduction

Theoretical DNA Computing is an area of biomolecular computing that loosely encompasses contributions to fundamental research in computer science originated in or motivated by research in DNA computing. One of the most active areas of research in theoretical DNA computing is the search for ways to encode information on DNA for the purposes of biocomputa-

tion that ensure that no unwanted bindings occur. The main premise is that information-encoding strings that are used in DNA computing experiments have an important property that differentiates them from their electronic computing counterparts. This property is the Watson-Crick complementarity between DNA single-strands that allows information-encoding strands to potentially interact.

Most DNA-based computations consist of three basic stages. The first is encoding the input data using DNA stranded molecules, the second is performing the biocomputation using bio-operations and the third is decoding the result. One of the main problems associated with such biocomputations is the design of the information-encoding oligonucleotides such that undesirable pairing due to the Watson-Crick complementarity is minimized. There are several approaches exist that address this sequence design problem. Such as the software simulation approach, the algorithmic approach and the theoretical approach to the design of optimal data-encoding DNA strands.

Theoretical DNA computing has been an enthusiastic research area in computer science for a decade. Studies include the computational properties of DNA recombination ([2]), the mathematical theory of DNA self-assembly ([1]), and the coding properties of DNA languages ([4]) has inspired the authors to study a similar subject. This study focus on a theoretical study of generalized notions of palindrome words. The motivation comes from the DNA encoded sequences which have an important property used in DNA

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computing. This property is the Watson-Crick complementarity between DNA single-strands. A single strand of DNA can be abstracted as a string sequence consisted of a combination of four different symbols. The four symbols are *adenine* (A), *guanine* (G), *cytosine* (C) and *thymine* (T). The Watson-Crick complement of a DNA strand satisfies two main properties, it is the reverse property and the complement property of the original strand. Mathematically, the Watson-Crick complementarity is translated into generalizing the identity involution. An involution is a function θ such that θ^2 equals the identity, that is, $\theta(A) = T$ and $\theta^2(A) = \theta(\theta(A)) = A$. Furthermore, for the DNA alphabet $\Delta = \{A, G, C, T\}$, an antimorphic involution is an involution θ with the additional property that $\theta(uv) = \theta(v)\theta(u)$ for all string sequences $u, v \in \Delta^*$, corresponds to the notion of the Watson-Crick complement of a DNA sequence.

2 Preliminaries

Assume X is an alphabet containing more than one letter. Let X^* be the free monoid generated by X . Every element of X^* is a *word* and every subset of X^* is a *language*. Let λ denote the empty word, and $X^+ = X^* \setminus \{\lambda\}$. For $w \in X^*$ and $L \subseteq X^*$, let $\lg(w)$ denote the length of the word w and let $|L|$ denote the cardinality of the language L . A language $L \subseteq X^*$ is *dense* if for any $w \in X^*$, there exist $x, y \in X^*$ such that $xwy \in L$. That is, for every $w \in X^*$, $X^*wX^* \cap L \neq \emptyset$. A *primitive word* is a word which is not a power of any other word. Let Q be the set of all primitive words over X . Every word $u \in X^+$ can be expressed as a power of a primitive word in a unique way, that is, for any $u \in X^+$, $u = f^n$ for a unique $f \in Q$ and $n \geq 1$. In this case, f is the primitive root of u and denoted by $\sqrt{u} = f$. Let $u = a_1a_2 \cdots a_n$ where $a_i \in X$. The reverse of the word u is $u^R = a_n \cdots a_2a_1$. A word u is called *palindrome* if $u = u^R$. Let R be the set of all palindrome words over X . The partial order relation $\leq_p$ (resp. $\leq_s$) is defined as: for $u, v \in X^*$, $v \leq_p u$ (resp. $v \leq_s u$) if and only if $u \in vX^*$ (resp. $u \in X^*v$). Moreover, the partial order relation $<_p$ (resp. $<_s$) is defined as: for $u, v \in X^+$, $v <_p u$ (resp. $v <_s u$) if and only if $u \in vX^+$ (resp. $u \in X^+v$).

Let $\theta : X^* \rightarrow X^*$ be a function such that $\theta^2 = I$ where I is the identity function. It can extend to a morphic involution on X^* if for all $u, v \in X^*$, $\theta(uv) = \theta(u)\theta(v)$ or an antimorphic involution if $\theta(uv) = \theta(v)\theta(u)$. We now recall some definitions

introduced by Kari and Mahalingam ([5]-[7]).

Definition 2.1 Let θ be either a morphic or an antimorphic involution on X^* .

(1) A word w is a θ -conjugate of another word u if $uv = \theta(v)w$ for some $v \in X^*$.

(2) A word $w \in X^*$ is called a θ -palindrome word if $w = \theta(w)$.

We also recall some observations on the above definitions. Let $\sigma(u)$ be the set containing all θ -conjugates of u . For example, let $X = \{a, b\}$ and θ be a morphic involution that maps a to b and vice versa. For $u = ababb$, $\sigma(u) = \{babaa, bbaba, bbbab, abbba, babbb, ababb\}$. Moreover, let θ be an antimorphic involution that maps a to b and vice versa. For $u = ababb$, $\sigma(u) = \{ababab, babab, bbbab, abbab, babbb, ababb\}$. Let R_θ be the set of all θ -palindrome words over X . Note that $aabb \in R_\theta$ where θ is antimorphic involution, but $aabb \notin R$. For any antimorphic involution θ , we give a definition of *skew θ -palindrome words* as following:

Definition 2.2 Let θ be an antimorphic involution on X^* . A word w is said to be *skew θ -palindrome* if $w = xy$ implies that $\theta(w) = yx$ for some $x, y \in X^*$.

Let θ is an antimorphic involution on X^* . Note that every θ -palindrome word is a skew θ -palindrome word because any palindrome word as a product of itself and the empty word λ . For example, $abab$ is θ -palindrome and is also a skew θ -palindrome word. Let $w = abba$. Then w is a skew θ -palindrome word, but it is not θ -palindrome.

Furthermore, we give some results which will be used in the rest of this study as follows:

Lemma 2.1(see [13]) Let $u, v \in X^+$. Then $uv = vu$ implies that u and v are powers of a common word.

Lemma 2.2(see [13]) If $u^m = v^n$ and $m, n \geq 1$, then u and v are powers of a common word.

3 Involution Palindrome Languages

A word u is called *palindrome* if it is the mirror image of itself. In this section we extend the concept of palindrome words to incorporate the

notion of involution function. The notion of θ -palindrome was defined in [6] and obtained independently in [8]. Note that if θ is the Watson-Crick involution, then the notion of Watson-Crick palindromes coincides with the term "palindrome" as used in molecular biology, especially in the study of enzymes.

The study of θ -palindromes for antimorphic involutions is interesting from two points of view: firstly, it may be desirable for certain DNA computing experiments to use DNA strands that contain θ -palindromic enzyme restriction sites as subwords, and secondly, in general, a set of DNA codewords should be free of θ -palindromic words, due to the intermolecular hybridizations that these would entail.

This study extends the understanding of involution palindrome words (θ -palindrome words). This notion is motivated by DNA strand design in the area of biocomputing where the Watson-Crick complementarity can be abstracted as an antimorphic involution function. A language consisting of involution palindrome words is defined as an involution palindrome language. Some properties of involution palindrome words and languages are surveyed in our study. Besides involution palindrome words being considered, some algebraic properties of skew involution palindrome words are studied as well.

Lemma 3.1 ([3]) Let θ be a morphic or an antimorphic involution. The word $f \in X^+$ is primitive if and only if $\theta(f)$ is primitive word.

Let $w \notin Q$. That is, $w = f^k$, $f \in Q$, $k \geq 2$. Then the primitive root of w is denoted by $\sqrt{w} = f$.

Lemma 3.2 ([3]) Let θ be a morphic or an antimorphic involution. Let $w \in X^+$. Then $\theta(\sqrt{w}) = \sqrt{\theta(w)}$.

For a language L , $L^{(i)} = \{w^i \mid w \in L\}$ for any $i \geq 1$. Then we have $Q^{(i)} = \{u^i \mid u \in Q\}$.

Lemma 3.3 ([7]) Let θ be a morphic or an antimorphic involution. For all $w \in X^+$, w is θ -palindrome if and only if $\sqrt{w}$ is θ -palindrome.

Lemma 3.4 ([7]) Let θ be an antimorphic involution. Then for $w \in X^+$, $w \in R_\theta$ if and only if $w = xy\theta(x)$, $x \in X^+$, $y \in X^*$ with $y \in R_\theta$.

Proposition 3.1 ([3]) Let θ be an antimorphic involution. Then $\sqrt{R_\theta} = R_\theta \cap Q$.

In the following propositions, we investigate the properties concerning R_θ^2 . When θ is an antimorphic involution, we show that $(R_\theta^2 \cap Q) \cap R_\theta = \emptyset$. Moreover, when θ is a morphic involution, $(R_\theta^2 \cap Q) \subseteq R_\theta$.

Proposition 3.2 ([3]) Let $w = uv \in Q$ with $u, v \in R_\theta$ where $u, v \in X^+$.

- (1) If θ is an antimorphic involution, then $w \notin R_\theta$.
- (2) If θ is a morphic involution, then $w \in R_\theta$.

Proposition 3.3 ([3]) Let θ be an antimorphic involution and $w = uv \in R_\theta \cap Q$ where $u, v \in X^+$. If $u \in R_\theta$, then $vu \notin R_\theta$.

A word $u \in X^*$ is a conjugate of $w \in X^*$ if there exists $v \in X^*$ such that $uv = vw$. In [6], authors have defined the θ -conjugate of a word for a morphic involution θ or an antimorphic involution θ as follow: $w \in X^*$ is a θ -conjugate of u such that $uv = \theta(v)w$ for some $v \in X^*$. We study the θ -conjugate of a word concerning θ -palindrome words in the following propositions.

Lemma 3.5 ([6]) Let w be a θ -conjugate of u . Then for a morphic involution θ , there exist $x, y \in X^*$ such that $u = xy$ and one of the following hold:

- (1) $w = y\theta(x)$ and $v = (\theta(x)\theta(y)xy)^i\theta(x)$ for some $i \geq 0$.
- (2) $w = \theta(y)x$ and $v = (\theta(x)\theta(y)xy)^i\theta(x)\theta(y)x$ for some $i \geq 0$.

Proposition 3.4 ([3]) Let $u \in X^+$ be a θ -palindrome word. Then for a morphic involution θ , any θ -conjugate of u is also θ -palindrome and is the product of two θ -palindrome words.

Lemma 3.6 ([6]) Let w be a θ -conjugate of u . Then for an antimorphic involution θ , there exist $x, y \in X^*$ such that either $u = xy$ and $w = y\theta(x)$ or $w = \theta(u)$.

Proposition 3.5 ([3]) Let $u \in X^+$ be a θ -palindrome word. Then for an antimorphic involution θ , any θ -conjugate $w \in X^+$ of u is one of the following:

- (1) w is also θ -palindrome.
- (2) $w \in \{\alpha\beta^2, \beta\alpha\beta\}$ with $\alpha \in R_\theta$ for some $\alpha \in X^*$, $\beta \in X^+$.

In the following propositions, some properties of the skew θ -palindrome words are studied when θ is an antimorphic involution.

Proposition 3.6 Let θ be an antimorphic involution on X^* . A word w is skew θ -palindrome if and only if w is a product of two θ -palindrome words.

Proof. Let θ be an antimorphic involution on X^* . If w is skew θ -palindrome, then there exist $u, v \in X^*$ such that $w = uv, \theta(w) = vu$. Since θ is an antimorphic involution, $\theta(w) = \theta(uv) = \theta(v)\theta(u)$. We have $\theta(v)\theta(u) = vu$. Hence $\theta(v) = v$ and $\theta(u) = u$. Then w is a product of two θ -palindrome words. Conversely, suppose that $w = uv$ where $u = \theta(u), v = \theta(v)$ for some $u, v \in X^*$. Then $\theta(w) = \theta(uv) = \theta(v)\theta(u) = vu$; hence w is a skew θ -palindrome word. ■

Proposition 3.6 is not true where θ is a morphic involution. For example, let $X = \{a, b\}$ and θ be a morphic involution that maps a to b and vice versa. As $w = ab, \theta(w) = \theta(ab) = \theta(a)\theta(b) = ba$. Then w is skew θ -palindrome. However, a, b are not θ -palindromes. Note that every θ -palindrome word is a skew θ -palindrome word because any palindrome word as a product of itself and the empty word λ .

Lemma 3.7([7]) Let θ be an antimorphic involution. Then for $u, v \in X^+, u, v \in R_\theta$ if and only if $(uv)^k u \in R_\theta$ for some $k \geq 0$.

Proposition 3.7 Let θ be an antimorphic involution. If w is skew θ -palindrome, then for $n \geq 2$, w^n has at least two different decompositions as product of two θ -palindromes.

Proof. Let θ be an antimorphic involution and w be a skew θ -palindrome word. By Proposition 3.6, there exist $w_1, w_2 \in R_\theta$ such that $w = w_1 w_2$. Then

$$\begin{aligned} w^n &= w_1((w_2 w_1)^{n-1} w_2) \\ &= (w_1 w_2 w_1)((w_2 w_1)^{n-2} w_2). \end{aligned}$$

Moreover, by Lemma 3.7, we have that $(w_1 w_2)^i w_1$ and $(w_2 w_1)^j w_2$ for $i, j \geq 0$ are θ -palindromes. This complete the proof. ■

Proposition 3.8 Let θ be an antimorphic involution. A word w is skew θ -palindrome if and only if w^n is skew θ -palindrome for $n \geq 2$.

Proof. Let θ be an antimorphic involution. If w is skew θ -palindrome, by Propositions 3.6 and 3.7,

then w^n is skew θ -palindrome for $n \geq 2$. Conversely, let w^n be a skew θ -palindrome word for $n \geq 2$. There exist $w_1, w_2 \in X^+$ with $w = w_1 w_2$ such that $w^n = (w^i w_1)(w_2 w^j)$ where $i + j = n - 1$ for some $i, j \geq 0$. By the definition of skew θ -palindrome word, we have $\theta(w^n) = (w_2 w^j)(w^i w_1)$. Since θ is an antimorphic involution,

$$\begin{aligned} \theta(w^n) &= \theta((w^i w_1)(w_2 w^j)) \\ &= \theta(w_2 w^j) \theta(w^i w_1) \\ &= (\theta(w))^j \theta(w_2) \theta(w_1) (\theta(w))^i \\ &= (\theta(w_2) \theta(w_1))^j \theta(w_2) (\theta(w_1) \theta(w_2))^i \theta(w_1) \\ &= (w_2 w^j)(w^i w_1). \end{aligned}$$

This implies that $\theta(w_1) = w_1$ and $\theta(w_2) = w_2$. By Proposition 3.6, w is skew θ -palindrome. ■

Given an involution θ , for any skew θ -palindrome word u , there exists a unique pair (x, y) such that $u = pq$ and $p = (xy)^{i-k-1}x$, $q = y(xy)^k$. We call (x, y) the twin-roots of u with respect to θ , or shortly θ -twin-roots of u .([12])

4 The Non-Involution Palindrome Words

In this section, we study the words which are not θ -palindrome for an antimorphic involution θ . To characterize the properties of non-involution palindrome words, we consider the θ -commutative relation which was defined in [6]. Let θ be either a morphic or an antimorphic involution. We recall that the θ -commutative relation in [6] is as follow: the θ -commutative relation $\leq_c^\theta$ on X^* is defined by

$$v \leq_c^\theta u \Leftrightarrow u = vx = \theta(x)v \text{ for some } x \in X^*, \text{ where } u, v \in X^*.$$

For $u \in X^*$, let $L_c^\theta(u) = \{v \in X^* | v \leq_c^\theta u\}$ and $N(u) = |L_c^\theta(u)|$. For $i \geq 1$, let $C_\theta(i) = \{u \in X^+ | N(u) = i\}$. For example, let $X = \{a, b\}$ and θ be an antimorphic involution that maps a to b and vice versa. Let $u = ab$. We have

- $u = ab \cdot 1 = \theta(1) \cdot ab$,
- $u = a \cdot b \neq \theta(b) \cdot a = aa$,
- $u = 1 \cdot ab = \theta(ab) \cdot 1 = \theta(b)\theta(a) = ab$.

Then $1, ab \in L_c^\theta(ab)$; hence $ab \in C_\theta(2)$. Note that the word ab is θ -palindrome for an antimorphic involution θ . However, the word $ab \in C_\theta(1)$

and is not θ -palindrome for a morphic involution θ .

In [6], authors observe that a word in $C_\theta(1)$ is not a θ -palindrome word when θ is an antimorphic involution. In the following lemma, we prove this observation in detail.

Lemma 4.1 ([3]) Let θ be an antimorphic involution. Then a word $u \in C_\theta(1)$ if and only if $u \notin R_\theta$.

For a word $u \in X^+$, let $\text{Pref}(u) = \{x \in X^+ \mid x \leq_p u\}$ and $\text{Suff}(u) = \{x \in X^+ \mid x \leq_s u\}$.

Proposition 4.1 ([3]) Let θ be an antimorphic involution and $u \in X^+$ be such that $\theta(\text{Pref}(u)) \cap \text{Suff}(u) = \emptyset$. Then $u \notin R_\theta$.

The converse of the statement in Proposition 4.1 does not hold in general. Let $X = \{a, b\}$ and θ be an antimorphic involution that maps a to b and vice versa. Let $u = aab$. Then $\theta(aab) = \theta(b)\theta(a)\theta(a) = abb \neq aab$, that is, $u \notin R_\theta$. But $b \in \theta(\text{Pref}(aab)) \cap \text{Suff}(aab)$.

Corollary 4.1 ([3]) Let θ be an antimorphic involution and $u \in X^+$ be such that $\theta(\text{Pref}(u)) \cap \text{Suff}(u) = \emptyset$. Then $u^+ \not\subseteq R_\theta$.

In the following proposition, we study the characteristic of non-involution palindrome word uv for any non-empty words u and v . First, we give a known result we need.

Lemma 4.2 ([6]) Let θ be an antimorphic involution and let $u, v \in C_\theta(1)$. If $\theta(\text{Pref}(u)) \cap \text{Suff}(v) = \emptyset$, then $uv \notin C_\theta(1)$.

Proposition 4.2 ([3]) Let θ be an antimorphic involution. Let $u, v \in X^+$ with $\theta(\text{Pref}(u)) \cap \text{Suff}(v) = \emptyset$. Then $uv \notin R_\theta$.

The converse of the statement in Proposition 4.2 does not hold in general. Let $X = \{a, b\}$ and θ be an antimorphic involution that maps a to b and vice versa. Let $u = aba$ and $v = ab$. Then $uv = abaab \neq \theta(uv) = abbab$; hence $uv \notin R_\theta$. However, from $\theta(\text{Pref}(u)) = \{b, ab, aba\}$ and $\text{Suff}(v) = \{b, ab\}$, we have $b, ab \in \theta(\text{Pref}(aba)) \cap \text{Suff}(ab)$; hence $\theta(\text{Pref}(u)) \cap \text{Suff}(v) \neq \emptyset$.

Corollary 4.2 ([3]) Let θ be an antimorphic involution. Let $u, v \in X^+$ with $\theta(\text{Pref}(u)) \cap \text{Suff}(v) = \emptyset$. Then $uv^k \notin R_\theta$ for any $k > 1$.

Corollary 4.3 ([3]) Let θ be an antimorphic involution and $u, v \in X^+$. If $\theta(\text{Pref}(u)) \cap \text{Suff}(v) = \emptyset$, then $u^k v \notin R_\theta$ for every $k > 1$.

Proposition 4.3 ([3]) Let θ be an antimorphic involution and $u, v \in X^+$. If $\theta(\text{Pref}(u)) \cap \text{Suff}(v) = \emptyset$, then $u^+ v^+ \not\subseteq R_\theta$.

In the following, we give a characterization for $u \notin R_\theta$ where $u \in X^+$ for $X = \{a, b\}$. We first need the following lemma.

Lemma 4.3 Let $X = \{a, b\}$ and θ be an antimorphic involution that maps a to b and vice versa. Let $u \in X^n$, $n \geq 1$. Let $u = u_1 u_2 \cdots u_n$, where $u_i \in X$ for every $1 \leq i \leq n$. Suppose that $u \in R_\theta$. Then $n = 2k$ for some $k \geq 1$ and $\theta(u_{k+1} \cdots u_{2k}) = u_1 \cdots u_k$.

Proof. Let $X = \{a, b\}$ and θ be an antimorphic involution that maps a to b and vice versa. Let $u \in X^n$, $n \geq 1$. Let $u = u_1 u_2 \cdots u_n$, where $u_i \in X$ for every $1 \leq i \leq n$. Suppose that $u \in R_\theta$. Then $\theta(u) = u$. That is, $\theta(u_n) \cdots \theta(u_1) = u_1 \cdots u_n$. If $n = 1$, then $u = a$ or $u = b$. That is, $\theta(u) = \theta(a) = b$ or $\theta(u) = \theta(b) = a$. Both contradict to $u \in R_\theta$. Hence $n \neq 1$. If $n = 2k + 1$ for some $k \geq 1$, then $\theta(u_{2k+1}) \cdots \theta(u_{k+2}) \theta(u_{k+1}) \theta(u_k) \cdots \theta(u_1) = u_1 \cdots u_k u_{k+1} u_{k+2} \cdots u_{2k+1}$. This implies that $\theta(u_{k+1}) = u_{k+1}$, a contradiction. Hence $n \neq 2k + 1$ for every $k \geq 1$. By above discussion, n is even. That is, $n = 2k$ for some $k \geq 1$. Now by the definition of θ -palindrome, we have $\theta(u_{k+1} \cdots u_{2k}) = u_1 \cdots u_k$. ■

Proposition 4.4 Let θ be an antimorphic involution that maps a to b and vice versa. Let $u \in X^n$, $n \geq 1$. Let $u = u_1 u_2 \cdots u_n$, where $u_i \in X$ for every $1 \leq i \leq n$. Then $u \notin R_\theta$ if and only if one of the following conditions holds:

- (1) n is odd;
- (2) n is even and there exists an integer $i < \frac{1}{2} \lg(u)$ such that $\theta(u_1 \cdots u_i) = u_{n-i+1} \cdots u_n$ and $\theta(u_1 \cdots u_i u_{i+1}) = u_{n-i} \cdots u_n$.

Proof. Let θ be an antimorphic involution. Let $u \in X^n$, $n \geq 1$. Let $u = u_1 u_2 \cdots u_n$, where $u_i \in X$ for every $1 \leq i \leq n$.

($\Leftarrow$) If $u \in R_\theta$, then by Lemma 4.3, $n = 2k$ for some $k \geq 1$ and $\theta(u_{k+1} \cdots u_{2k}) = u_1 \cdots u_k$. Since $n = 2k$ for some $k \geq 2$, (1) is not true. From $\theta(u_{k+1} \cdots u_{2k}) = u_1 \cdots u_k$, we have that $\theta(u_{2k}) \cdots \theta(u_{k+1}) = u_1 \cdots u_k$. This implies that (2) is not true.

($\Rightarrow$) Clearly. ■

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